

# 1 PRELIMINARIES

We study behavior cloning from demonstrations  $\mathcal{D} = \{(o_i, a_i)\}$ , where  $o_i$  contains the observations and task instruction, and  $a_i$  is an action chunk. The goal is to learn a policy that generates action chunks conditioned on these inputs to accomplish the demonstrated tasks. A standard baseline is direct action regression (Mandlekar et al., 2021), which trains a network  $f_\theta(o)$  to predict an action chunk by minimizing the mean squared error (MSE):

$$\mathcal{L}_{\text{MSE}}(\theta) = \mathbb{E}_{(o,a) \sim \mathcal{D}} [\|f_\theta(o) - a\|_2^2]. \quad (1)$$

However, denoising-based policies have outperformed conventional behavior-cloning baselines on challenging manipulation tasks (Chi et al., 2023; Pan et al., 2026). A common explanation attributes this advantage to multimodal demonstrations: generative policies can represent multiple valid actions for the same observation, whereas direct MSE regression can average incompatible actions (Chi et al., 2023).

## 1.1 MULTIMODALITY AND DETERMINISTIC POLICIES

Pan et al. (2026) challenge this explanation. In their Flow-policy probes, action samples form a single cluster even at symmetry-critical or ambiguous states, and executing the mean sampled action has little effect on task success. The Flow-regression gap also persists on data collected from a deterministic expert, indicating that multimodality does not explain the performance gap.

Deterministic regression can match Flow’s performance. The minimal iterative policy (MIP) of Pan et al. (2026), trained with squared-error supervision and noise injection, generates actions deterministically in two network evaluations. Across seven challenging tasks, the authors report an average success rate relative to Flow of 1.02 for MIP and 0.74 for direct regression (their Figure 1). These results show that modeling multiple action modes is not necessary for strong control performance: regression-based policies can match Flow.

## 1.2 M-ESTIMATION AND ROBUST REGRESSION

M-estimation generalizes least squares by replacing the squared error with a residual penalty (Huber, 1964). For the continuous action channels, let  $r_\theta = f_\theta(o) - a \in \mathbb{R}^d$  denote the flattened action-chunk residual. An M-estimator minimizes

$$\mathcal{L}_\rho(\theta) = \mathbb{E}_{(o,a) \sim \mathcal{D}} [\rho(r_\theta)]. \quad (2)$$

Choosing  $\rho(r) = -\log q(r)$  for a residual density  $q$  gives maximum likelihood estimation (MLE). In particular, an isotropic Gaussian  $q = \mathcal{N}(0, \sigma^2 I_d)$  with fixed  $\sigma > 0$  yields the MSE objective in Eq. 1, up to a positive multiplicative factor and an additive constant.

Robust regression reduces sensitivity to extreme residuals through the shape of  $\rho$ . The prediction gradient  $\psi(r) = \nabla_r \rho(r)$  equals  $2r$  for squared error and grows linearly with residual magnitude. By contrast, coordinate-wise Huber penalties are quadratic near zero and linear beyond a threshold, yielding bounded derivatives (Huber, 1964).

## 1.3 HETEROSCEDASTIC REGRESSION

Heteroscedastic regression allows the conditional residual scale to vary with the input. A common formulation jointly predicts an action mean  $f_\theta(o)$  and a positive scalar scale  $\sigma_\theta(o)$ , using the Gaussian model  $p_\theta(a | o) = \mathcal{N}(a; f_\theta(o), \sigma_\theta(o)^2 I_d)$  (Nix & Weigend, 1994; Seitzer et al., 2022). Omitting additive constants, its negative log-likelihood is

$$\mathcal{L}_{\text{HG}}(\theta) = \mathbb{E}_{(o,a) \sim \mathcal{D}} \left[ \frac{\|r_\theta\|_2^2}{2\sigma_\theta(o)^2} + d \log \sigma_\theta(o) \right]. \quad (3)$$

Inverse-variance weighting measures each residual relative to its predicted local scale, while the logarithmic term penalizes scale inflation. This adapts the scale across inputs but retains a quadratic penalty on the standardized residual  $r_\theta/\sigma_\theta(o)$ .

Together, these formulations distinguish input-dependent scale from sensitivity to extreme residuals, while retaining deterministic inference through  $f_\theta(o)$ . We next examine both properties in the demonstration data.

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